

Kubernetes Assignment

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1. L1 - Create Deployment Controller Object to Deploy the Application Image Created in Docker Module and Expose it to the Internet.

The screenshot shows the AWS Management Console interface for an EC2 instance. The breadcrumb navigation at the top reads 'EC2 > Instances > i-09b11d304f24b0fc5'. The main heading is 'Instance summary for i-09b11d304f24b0fc5 (Docker Server)'. Below this are buttons for 'Connect', 'Instance state', and 'Actions'. A status bar indicates 'Updated 13 minutes ago' and 'Running'. The instance details are organized into three columns:

Instance ID	Public IPv4 address	Private IPv4 addresses
i-09b11d304f24b0fc5	16.171.238.110 open address	172.31.15.198

IPv6 address	Instance state	Public DNS
-	Running	ec2-16-171-238-110.eu-north-1.compute.amazonaws.com open address

Hostname type	Private IP DNS name (IPv4 only)	Elastic IP addresses
IP name: ip-172-31-15-198.eu-north-1.compute.internal	ip-172-31-15-198.eu-north-1.compute.internal	-

Answer private resource DNS name	Instance type	AWS Compute Optimizer finding
-	t3.micro	Opt-in to AWS Compute Optimizer for recommendations. Learn more

Auto-assigned IP address	VPC ID	
16.171.238.110 [Public IP]	vpc-04fbb5d5ae5f66a87	

- Create an IAM user to communicate with the Kubernetes cluster.

The screenshot shows the 'Create user' wizard in the AWS IAM console, specifically the 'Retrieve password' step. A green success message at the top states 'User created successfully' and provides a 'View user' button. A progress indicator on the left shows four steps: 'Specify user details', 'Set permissions', 'Review and create', and 'Retrieve password' (which is the current step). The main content area is titled 'Retrieve password' and includes a 'Console sign-in details' box with the following information:

- Console sign-in URL: <https://787169320097.signin.aws.amazon.com/console>
- User name: eks-admin
- Console password: [masked] [Show](#)

At the bottom, there are three buttons: 'Cancel', 'Download .csv file', and 'Return to users list'.

☰ IAM > Users > eks-admin > Create access key

Step 1

● Access key best practices & alternatives

Step 2 - optional

● Set description tag

Step 3

● **Retrieve access keys**

Retrieve access keys [Info](#)

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
 AKIA3ORXSTSQW52E5FVL	 ***** Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

[Download .csv file](#) [Done](#)

- Install the AWS CLI and configure it with the required credentials.

```
[ec2-user@ip-172-31-15-198 ~]$ aws --version
aws-cli/2.33.12 Python/3.13.11 Linux/6.1.159-182.297.amzn2023.x86_64 exe/x86_64.amzn.2023
[ec2-user@ip-172-31-15-198 ~]$ aws configure
AWS Access Key ID [None]: AKIA3ORXSTSQW52E5FVL
AWS Secret Access Key [None]: 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
Default region name [None]: eu-north-1
Default output format [None]: json
[ec2-user@ip-172-31-15-198 ~]$ aws sts get-caller-identity
{
  "UserId": "AIDA3ORXSTSQ5RIZKGCRW",
  "Account": "787169320097",
  "Arn": "arn:aws:iam::787169320097:user/eks-admin"
}
[ec2-user@ip-172-31-15-198 ~]$
```

- Install `kubectl` and `eksctl` to manage Kubernetes clusters on AWS.

```
[ec2-user@ip-172-31-15-198 ~]$ kubectl version --client
Client Version: v1.29.0-eks-5e0fdde
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
[ec2-user@ip-172-31-15-198 ~]$ eksctl version
0.221.0
[ec2-user@ip-172-31-15-198 ~]$
```

- Create a cluster named **my-ekscluster1** with one node in the same region as the EC2 machine.

```

[ec2-user@ip-172-31-15-198 ~]$ eksctl create cluster \
--name my-ekscluster1 \
--region eu-north-1 \
--nodegroup-name my-nodes \
--nodes 1 \
--node-type t3.small \
--managed
2026-02-03 08:23:36 [i] eksctl version 0.221.0
2026-02-03 08:23:36 [i] using region eu-north-1

```

```

[ec2-user@ip-172-31-15-198 ~]$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-192-168-5-86.eu-north-1.compute.internal Ready    <none>   12m   v1.32.9-eks-ecaa3a6

```

- Log in to Docker Hub and push the tagged application image.

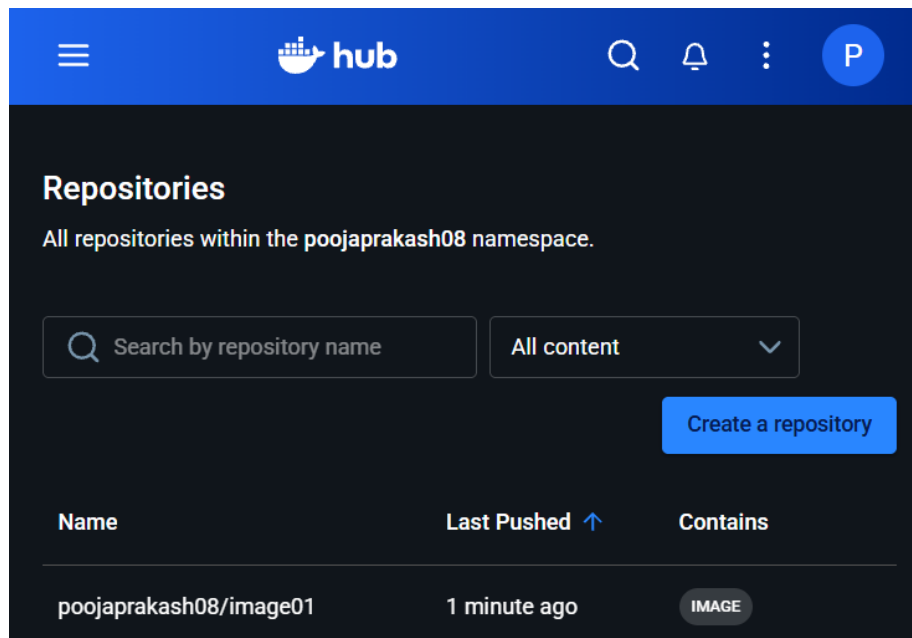
```

[ec2-user@ip-172-31-15-198 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
image01       v1        0dc11ba087e4   22 seconds ago 140MB
<none>        <none>    2d4389d8910f   41 hours ago   140MB
mysql         8.0       67fcbe35dfe8   3 days ago     786MB
[ec2-user@ip-172-31-15-198 ~]$ docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/

Username: poojaprakash08
Password:
WARNING! Your password will be stored unencrypted in /home/ec2-user/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[ec2-user@ip-172-31-15-198 ~]$ docker tag image01:v1 poojaprakash08/image01:v2
[ec2-user@ip-172-31-15-198 ~]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
image01       v1        0dc11ba087e4   About a minute ago 140MB
poojaprakash08/image01 v2        0dc11ba087e4   About a minute ago 140MB
<none>        <none>    2d4389d8910f   41 hours ago     140MB
mysql         8.0       67fcbe35dfe8   3 days ago       786MB
[ec2-user@ip-172-31-15-198 ~]$ docker push poojaprakash08/image01:v2
The push refers to repository [docker.io/poojaprakash08/image01]
05c3ab9b6383: Pushed
9c6a45fc5ba1: Pushed
b028c344a975: Pushed
5c43b6d1c516: Pushed
06f252d15777: Mounted from library/python
072aa7e4d8cf: Mounted from library/python
5662d93ce7b0: Mounted from library/python
a8ff6f8cbdfd: Mounted from library/python
v2: digest: sha256:8da4ffeeabb526d402f303947b5044841547513e9e1813f8a8888391335bc387 size: 1990
[ec2-user@ip-172-31-15-198 ~]$

```



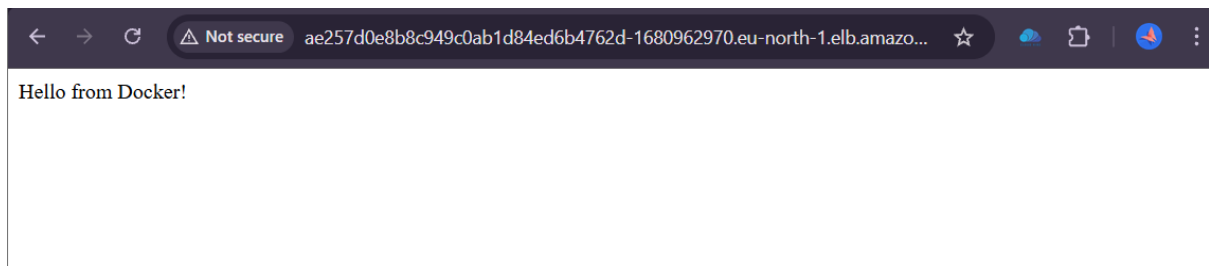
- Write a YAML file to define the Deployment and Service for the application.

```
! myapp.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: myapp-deployment
5  spec:
6    replicas: 2
7    selector:
8      matchLabels:
9        app: myapp
10   template:
11     metadata:
12       labels:
13         app: myapp
14     spec:
15       containers:
16       - name: myapp-container
17         image: poojaprakash08/image01:v2
18         ports:
19         - containerPort: 80
20   ---
21   apiVersion: v1
22   kind: Service
23   metadata:
24     name: myapp-service
25   spec:
26     type: LoadBalancer
27     selector:
28       app: myapp
29     ports:
30     - port: 80
31       targetPort: 80
32
```

- Apply the YAML file, which creates the Deployment, Pods, and Service objects.

```
[ec2-user@ip-172-31-15-198 ~]$ vi myapp.yml
[ec2-user@ip-172-31-15-198 ~]$ kubectl apply -f myapp.yml
deployment.apps/myapp-deployment created
service/myapp-service created
[ec2-user@ip-172-31-15-198 ~]$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
myapp-deployment    2/2     2             2           15s
[ec2-user@ip-172-31-15-198 ~]$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
myapp-deployment-64f5dfb5cb-6zjzp  1/1     Running   0           28s
myapp-deployment-64f5dfb5cb-ns6cn  1/1     Running   0           28s
[ec2-user@ip-172-31-15-198 ~]$ kubectl get services
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP
kubernetes          ClusterIP   10.100.0.1       <none>
myapp-service       LoadBalancer 10.100.100.211   ae257d0e8b8c949c0ab1d84ed6b4762d-1680962970.eu-north-1.elb.amazonaws.com
[ec2-user@ip-172-31-15-198 ~]$ kubectl get svc myapp-service
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP
myapp-service       LoadBalancer 10.100.100.211   ae257d0e8b8c949c0ab1d84ed6b4762d-1680962970.eu-north-1.elb.amazonaws.com
[ec2-user@ip-172-31-15-198 ~]$
```

- Access the application using the external IP address provided by the Service.



- An AWS EKS cluster was successfully created to run Kubernetes workloads. A Deployment controller was used to manage multiple replicas of the Dockerized application. The application was exposed to the internet using a LoadBalancer Service, which automatically provisioned an AWS Elastic Load Balancer.

2. L2 - Scale-up and Scale-Down the Pods Deployed

- Scale the Deployment from 2 pods to 3 pods using the `kubectl scale` command.

```
[ec2-user@ip-172-31-15-198 ~]$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
myapp-deployment    2/2     2             2           35m
[ec2-user@ip-172-31-15-198 ~]$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myapp-deployment-5dc95f57d5-6wkfz   1/1     Running   0          27m
myapp-deployment-5dc95f57d5-qwxq9   1/1     Running   0          27m
[ec2-user@ip-172-31-15-198 ~]$ kubectl scale deployment myapp-deployment --replicas=3
deployment.apps/myapp-deployment scaled
[ec2-user@ip-172-31-15-198 ~]$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myapp-deployment-5dc95f57d5-4fxtq   1/1     Running   0          14s
myapp-deployment-5dc95f57d5-6wkfz   1/1     Running   0          28m
myapp-deployment-5dc95f57d5-qwxq9   1/1     Running   0          28m
[ec2-user@ip-172-31-15-198 ~]$
```

- Scale down the Deployment, reducing the number of pods to 1

```
[ec2-user@ip-172-31-15-198 ~]$ kubectl scale deployment myapp-deployment --replicas=1
deployment.apps/myapp-deployment scaled
[ec2-user@ip-172-31-15-198 ~]$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myapp-deployment-5dc95f57d5-4fxtq   1/1     Terminating   0          104s
myapp-deployment-5dc95f57d5-6wkfz   1/1     Running        0          30m
myapp-deployment-5dc95f57d5-qwxq9   1/1     Terminating   0          30m
[ec2-user@ip-172-31-15-198 ~]$
```

- Using the Kubernetes Deployment controller, the application Pods were dynamically scaled up and down with the `kubectl scale` command. Kubernetes automatically created and terminated Pods while maintaining service availability.

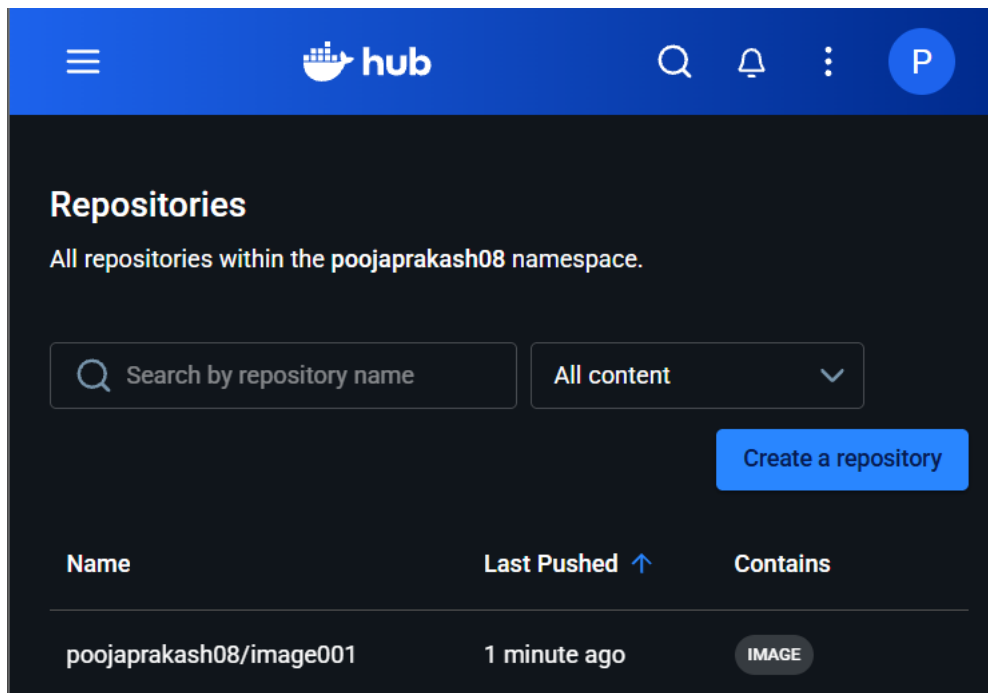
3. L3 - Implement Rolling-Update Strategy to Upgrade the Application Image from V1.0 to V1.1

- Modify the `app.py` file to introduce visible changes for the rolling update.
- Build a new Docker image with the updated files.

```
[root@ip-172-31-15-198 Docker_Assignment]# ls
Dockerfile app.py docker-compose.yml requirements.txt
[root@ip-172-31-15-198 Docker_Assignment]# vi app.py
[root@ip-172-31-15-198 Docker_Assignment]# docker build -t image002:v1 .
[+] Building 0.6s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 259B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 0.4s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:d0d43a8b0c352c215cd1381f3f4d7ac34cf3440cd041 0.0s
=> => resolve docker.io/library/python:3.11-slim@sha256:d0d43a8b0c352c215cd1381f3f4d7ac34cf3440cd041 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 558B                                  0.0s
=> CACHED [2/5] WORKDIR /app                                    0.0s
=> CACHED [3/5] COPY requirements.txt .                          0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY app.py .                                         0.0s
=> exporting to image                                           0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:0e71214660884bf7c6e4ada5f72466ee38d46ce5274211793419baa9391053f3 0.0s
=> => naming to docker.io/library/image002:v1                  0.0s
[root@ip-172-31-15-198 Docker_Assignment]# docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
image002            v1          0e7121466088     5 seconds ago   140MB
image01             v1          0dc11ba087e4     2 hours ago     140MB
poojaprakash08/image01 v2          0dc11ba087e4     2 hours ago     140MB
<none>              <none>      2d4389d8910f     42 hours ago    140MB
mysql               8.0         67fcbe35dfe8     3 days ago      786MB
[root@ip-172-31-15-198 Docker_Assignment]#
```

- Tag the new Docker image and push it to Docker Hub.

```
[root@ip-172-31-15-198 Docker_Assignment]# docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
image002            v1          0e7121466088   5 seconds ago  140MB
image01             v1          0dc11ba087e4   2 hours ago    140MB
poojaprakash08/image01 v2          0dc11ba087e4   2 hours ago    140MB
<none>              <none>      2d4389d8910f   42 hours ago   140MB
mysql               8.0         67fcbe35dfe8   3 days ago     786MB
[root@ip-172-31-15-198 Docker_Assignment]# docker tag image002:v1 poojaprakash08/image001:v1.1
[root@ip-172-31-15-198 Docker_Assignment]# docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
image002            v1          0e7121466088   About a minute ago  140MB
poojaprakash08/image001 v1.1        0e7121466088   About a minute ago  140MB
image01             v1          0dc11ba087e4   2 hours ago     140MB
poojaprakash08/image01 v2          0dc11ba087e4   2 hours ago     140MB
<none>              <none>      2d4389d8910f   42 hours ago    140MB
mysql               8.0         67fcbe35dfe8   3 days ago      786MB
[root@ip-172-31-15-198 Docker_Assignment]# docker push poojaprakash08/image001:v1.1
The push refers to repository [docker.io/poojaprakash08/image001]
66a0c49b144e: Pushed
9c6a45fc5ba1: Mounted from poojaprakash08/image01
b028c344a975: Mounted from poojaprakash08/image01
5c43b6d1c516: Mounted from poojaprakash08/image01
06f252d15777: Mounted from poojaprakash08/image01
072aa7e4d8cf: Mounted from poojaprakash08/image01
5662d93ce7b0: Mounted from poojaprakash08/image01
a8ff6f8cbdfd: Mounted from poojaprakash08/image01
v1.1: digest: sha256:4a760ce535313fb667810fd56209dd5d4169ef5e0091b8fa9a2ac3cd70b47b84 size: 1990
[root@ip-172-31-15-198 Docker_Assignment]#
```



The screenshot shows the Docker Hub web interface. At the top is a blue navigation bar with the Docker Hub logo, a search icon, a bell icon, a menu icon, and a profile icon labeled 'P'. Below the navigation bar, the page title 'Repositories' is displayed, followed by the subtitle 'All repositories within the poojaprakash08 namespace.'.

Below the subtitle, there is a search bar with the placeholder text 'Search by repository name' and a dropdown menu currently set to 'All content'. To the right of these is a blue button labeled 'Create a repository'.

The main content area displays a table of repositories. The table has three columns: 'Name', 'Last Pushed', and 'Contains'. The first row shows the repository 'poojaprakash08/image001' with a 'Last Pushed' time of '1 minute ago' and a 'Contains' button labeled 'IMAGE'.

Name	Last Pushed	Contains
poojaprakash08/image001	1 minute ago	IMAGE

- Update the Deployment YAML file to reference the new image version.

```

app: myapp
strategy:
  rollingUpdate:
    maxSurge: 25%
    maxUnavailable: 25%
  type: RollingUpdate
  rollingUpdate:
    maxSurge: 1
    maxUnavailable: 0
template:
  metadata:
    creationTimestamp: null
    labels:
      app: myapp
  spec:
    containers:
      - image: poojaprakash08/image001:v1.1
        imagePullPolicy: IfNotPresent
        name: myapp-container
        ports:
          - containerPort: 5000
            protocol: TCP

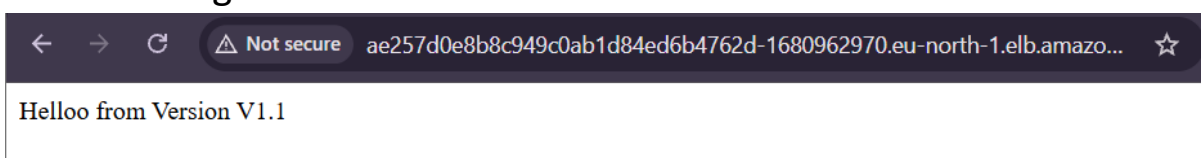
```

```

[ec2-user@ip-172-31-15-198 ~]$ kubectl edit deployment myapp-deployment
deployment.apps/myapp-deployment edited
[ec2-user@ip-172-31-15-198 ~]$ kubectl describe deployment myapp-deployment
Name:                myapp-deployment
Namespace:            default
CreationTimestamp:    Tue, 03 Feb 2026 09:34:04 +0000
Labels:               <none>
Annotations:          deployment.kubernetes.io/revision: 3
Selector:              app=myapp
Replicas:              3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:          RollingUpdate
MinReadySeconds:       0
RollingUpdateStrategy: 0 max unavailable, 1 max surge
Pod Template:
  Labels:  app=myapp
  Containers:
    myapp-container:
      Image:          poojaprakash08/image001:v1.1
      Port:           5000/TCP
      Host Port:      0/TCP

```

- Access the application again using the same LoadBalancer URL to verify the changes.



- A rolling update strategy was implemented using a Kubernetes Deployment to upgrade the application image from version v1.0 to v1.1. Kubernetes gradually replaced old Pods with new Pods while ensuring zero downtime. The update and rollback process was managed using ``kubectl rollout`` commands.